



# Temperature-dependent lithium storage behavior in tetragonal boron (B<sub>50</sub>) thin film anode for Li-ion batteries

Xingle Ding<sup>a,c,1</sup>, Xia Lu<sup>b,1</sup>, Zhengwen Fu<sup>c,\*</sup>, Hong Li<sup>b,\*</sup>

<sup>a</sup> Shanghai Key laboratory of Molecular Catalysts and Innovative Materials, State Key Laboratory of Surface Physics & Department of Physics, Fudan University, Shanghai 200433, China

<sup>b</sup> Beijing National Laboratory for Condensed Matter Physics, Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China

<sup>c</sup> Department of Chemistry & Laser Chemistry Institute, Fudan University, Shanghai 200433, China

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## ABSTRACT

Boron (B) has a high theoretical lithium storage capacity of 3046 mAh g<sup>-1</sup> for forming Li<sub>5</sub>B<sub>4</sub> phase. Li–B alloy has been used in high temperature thermal lithium batteries. In this work, a new tetragonal boron (B<sub>50</sub>) thin film with a thickness of 80 nm has been deposited on a vanadium coated glass substrate by a pulse laser deposition (PLD) method. It is found that this electrode film shows only a lithium storage capacity of 43 mAh g<sup>-1</sup> in a nonaqueous electrolyte at room temperature. According to a first-principles calculation, the B<sub>50</sub> is a metallic conductor. Therefore, the very poor activity could be related to poor lithium ion diffusion property since the diffusion barrier in this tetragonal B<sub>50</sub> lattice is calculated as 2.59 eV. This consists well with the fact that the lithium storage capacity in the B<sub>50</sub> thin film is improved significantly to 268 mAh g<sup>-1</sup> at 85 °C.

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## 1. Introduction

There are at least four major polymorphs ( $\alpha$ ,  $\beta$ ,  $\gamma$  and T) existing for B element, in which band gap of  $\alpha$ ,  $\beta$  and  $\gamma$  phases is 2.0 eV, 1.6 eV and 2.1 eV, respectively [1,2]. All the phases, even disordered (amorphous) boron, contain crystallography regular boron icosahedra B<sub>12</sub> [3–5].

Theoretical specific gravimetric capacity of lithium storage in B is 2846 mAh g<sup>-1</sup> for forming the Li<sub>7</sub>B<sub>6</sub> phase [6], 3046 mAh g<sup>-1</sup> for forming Li<sub>5</sub>B<sub>4</sub> alloy (R<sub>3m</sub>, rhombohedral phase, 85–0553) [7] or 4873 mAh g<sup>-1</sup> for forming Li<sub>2</sub>B phase [8] or 7309 mAh g<sup>-1</sup> for forming Li<sub>3</sub>B phase [9] in term of the weight of boron. The electrochemical reactivity of boron with lithium is very poor at room temperature. In the electrolyte of anhydrous 1 M LiPF<sub>6</sub> in a 1:1 mixture of ethylene carbonate (EC) and dimethyl carbonate (DMC) (by volume), Zhou et al. found that LiB compound showed a reversible capacity of 293 mAh g<sup>-1</sup> at a voltage range of 0.0–0.75 V vs. Li<sup>+</sup>/Li [10]. In contrast, Li–B alloy showed high reactivity in molten salt electrolyte at high temperature (673–773 K) in lithium thermal batteries [11,12]. Therefore, the limited room-temperature (RT)

reactivity of boron may be related to its poor kinetic property, which has not been clarified clearly.

It is doubt whether the reactivity of boron at room temperature can be enhanced by the size effect. In this paper, a B thin film with a thickness of 80 nm has been fabricated by a pulsed laser deposition (PLD) method for the first time. The electrochemical tests indicate that the specific capacity of the B thin film is very low at room temperature but can be enhanced strongly at 85 °C. The origin of the poor kinetic property of lithium storage in the B<sub>50</sub> lattice is studied by a first-principles calculation.

## 2. Experimental and simulation

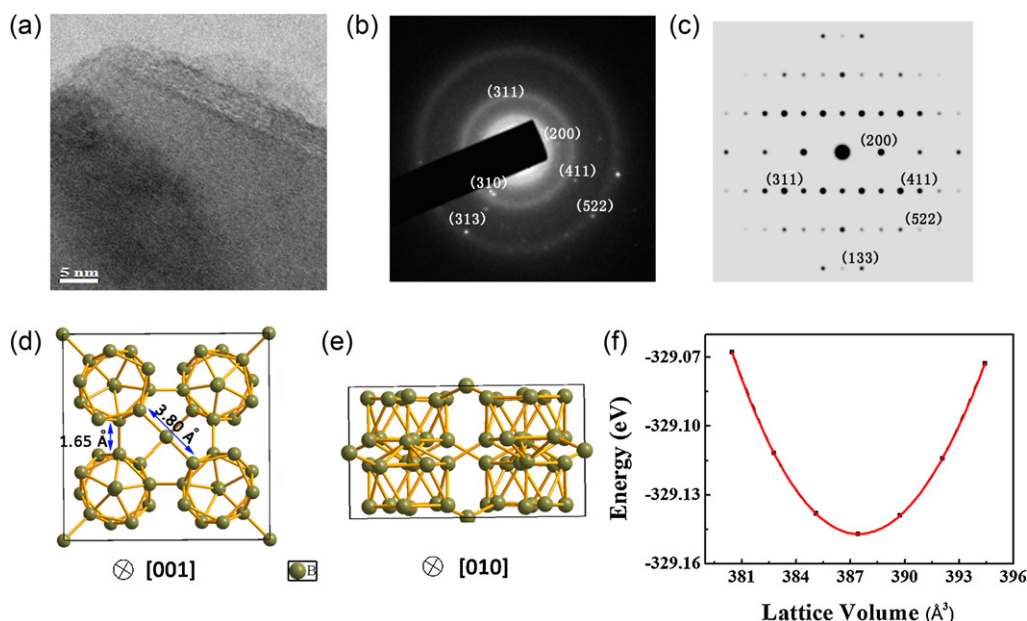
### 2.1. Experimental

The PLD system used in this work was described previously [13]. A 355 nm laser beam from a Q-switched Nd: yttrium aluminum garnet laser (Spectra Physics GCR-150) with a pulsed repetition rate of 10 Hz and a pulsed width of 5 ns was focused on the surface of the target. The incident angle between the laser beam and the target surface was 45°. The laser intensity was about 2 J cm<sup>-2</sup>. The distance between the target and the substrate was ca. 4 cm. The sputtering chamber was evacuated below 10<sup>-2</sup> Pa during deposition. Pure boron target was obtained by cold pressing boron powder (99%, Alpha). The film was deposited on a vanadium coated glass substrate at room temperature. The thickness of the film was

\* Corresponding authors. Tel.: +86 10 82648067; fax: +86 10 82649046.

E-mail addresses: [zwfu@fudan.edu.cn](mailto:zwfu@fudan.edu.cn) (Z. Fu), [hli@iphy.ac.cn](mailto:hli@iphy.ac.cn) (H. Li).

<sup>1</sup> Xingle Ding and Xia Lu contributed equally to this work.



**Fig. 1.** (a) HRTEM image of as-prepared  $B_{50}$  thin film; (b) selected area electron diffraction (SAED) images; (c) simulated SAED at [0 1 1] zone axis; schematic views of tetragonal  $B_{50}$  lattice: (d) top view and (e) side view; (f) energy varies with the change of  $B_{50}$  lattice volume by fitting to the Birch–Murnaghan equation of state [22].

about 80 nm for 30 min deposition. High resolution transmission electron microscopy (HRTEM) and selected area electron diffraction (SAED) investigations were carried out by a JEOL 2010 TEM microscope at 160 kV accelerating voltage. An energy-dispersive X-ray analysis (EDX, Kevex EDX spectrometer) coupled with the microscope was used to determine the chemical composition of the products. XPS measurements were performed with a PerkinElmer PHI 6000CESCA system with monochromatic Al  $K\alpha$  (1486.6 eV) irradiation. To correct possible charging of the films by X-ray irradiation, the binding energy was calibrated using C 1s (284.6 eV) spectrum of hydrocarbon that remained in the XPS analysis chamber.

Three-electrodes, the as-deposited  $B_{50}$  thin film as a working electrode and two lithium sheets as a counter electrode and a reference electrode, were used to perform discharging and charging measurements between 0.01 V and 3.0 V vs.  $Li^+/Li$  at constant current density. The electrolyte is 1 M  $LiPF_6$  in ethylene carbonate and dimethyl carbonate with a volume ratio of 1:1 (Merck). Charge–discharge measurements were performed using a Land BT 1-40 battery test system. The battery fabrication was done in an Ar-filled glove box and the electrode materials were rinsed in anhydrous DMC to eliminate residual salts before analysis.

## 2.2. First-principles simulation

All the first-principles calculations were performed by the Vienna *ab initio* simulation package (VASP) [14,15]. The valence and core interactions were studied by the projector augmented-wave (PAW) method within the generalized gradient approximation (GGA) [16,17], based on the density functional theory (DFT) with spin polarizations. The gradient-corrected

Perdew–Burke–Ernzerhof (PBE) [18,19] was employed for calculating the electron exchange correlation energy. The configurations  $Li\ 2s^1$  and  $B\ 2s^2 2p^1$  were treated as the valence electrons. The irreducible Brillouin zone integration was performed using the  $4 \times 4 \times 4$  Monkhorst–Pack k-points grid [20] with a Gaussian smearing of 0.1 eV near the Fermi level ( $E_F$ ). Energy cut-off for plane-wave and density were chosen to be 500 eV. The convergence error of the total energy is about  $10^{-5}$  eV (per primitive cell). Atomic sites and lattice constants were relaxed until the residual forces were less than 0.01 eV/Å in each species.

## 3. Results and discussion

### 3.1. Lattice of tetragonal $B_{50}$

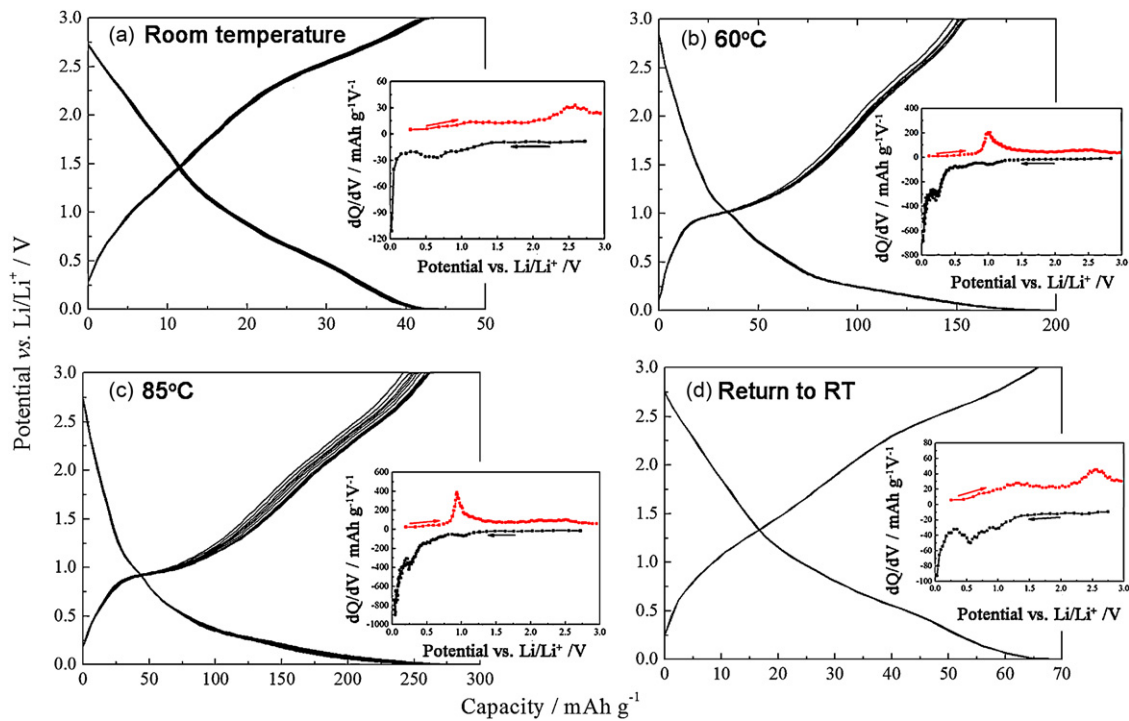
XRD measurements failed to identify the structure of as-deposited  $B_{50}$  thin films. The HRTEM and SAED images of the as-prepared  $B_{50}$  thin film are shown in Fig. 1(a) and (b). Some clear rings in Fig. 1(b) indicate the nanosized polycrystalline nature of the as-deposited film. All the measured  $d$ -spacings from the SAED image can be assigned to these of the tetragonal boron, which is also in consistency with the simulated SAED image at [0 1 1] zone axis in Fig. 1(c).

Tetragonal boron lattice presents a  $P4_2/nmm$  space group (group index: 134), with 50 boron atoms accommodated in the unit cell in which each boron icosahedra ( $B_{12}$ ) forms a caged-structure and four cages are connected by the rest two B atoms [21]. Fig. 1(d) and (e) show the top [0 0 1] and side [0 1 0] views of the  $B_{50}$  unit cell. The distance of the adjacent cage is ca. 1.65 Å and the diagonal distance of the cages is ca. 3.80 Å. There is a large empty

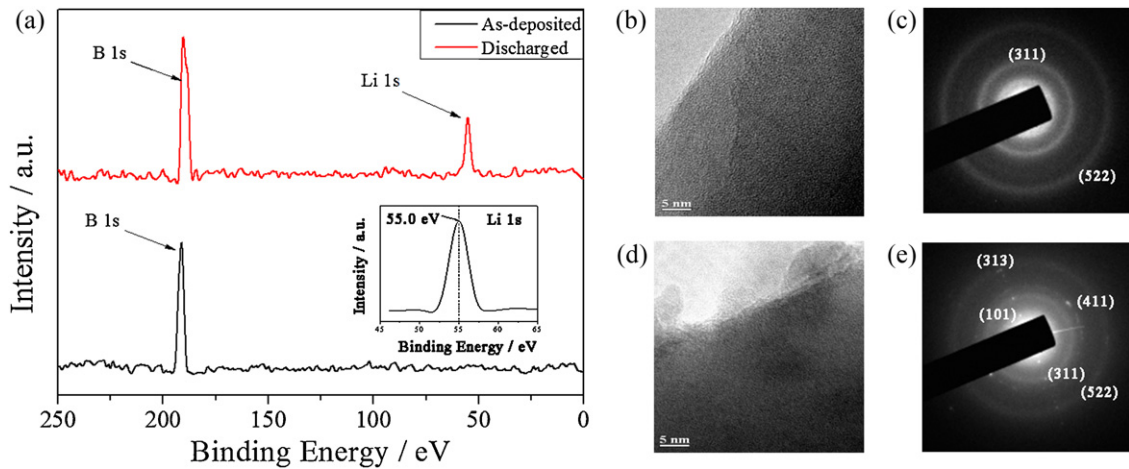
**Table 1**

Calculated lattice parameters ( $a$  and  $c$ ), volume of the unit cell ( $V_0$ ), bulk modulus ( $B_0$ ) and differential of the  $B_0$  to the volume ( $B'$ ) for tetragonal B lattice. The experimental data for tetragonal B are  $a = 8.75$  Å and  $c = 5.06$  Å taken from Ref. [17].

| Parameters | $F_0$ (eV) | $B_0$ (GPa) | $V_0$ (Å <sup>3</sup> ) | $B'$    | $a$ (Å) | $c$ (Å) |
|------------|------------|-------------|-------------------------|---------|---------|---------|
| Values     | −329.14    | 195.76      | 387.46                  | 3.26675 | 8.851   | 4.945   |
| Errors     | 2.5E−4     | 0.0777      | 0.002                   | 0.12319 | 1.15%   | −2.09%  |



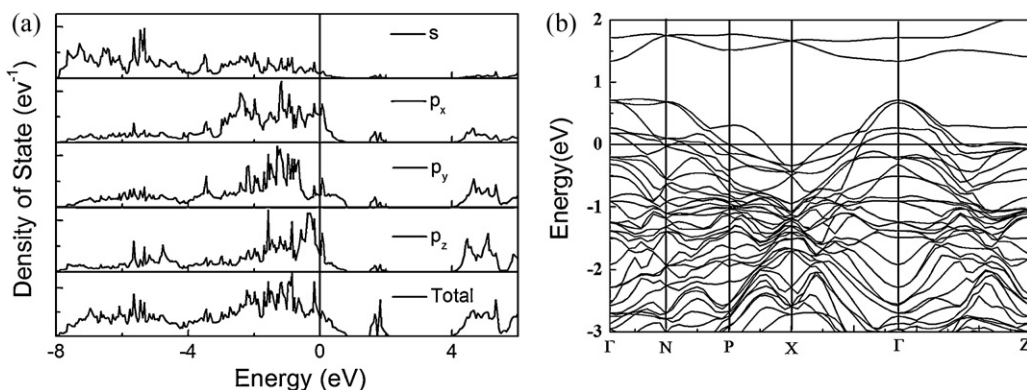
**Fig. 2.** The galvanostatic discharge/charge voltage curves with differential capacitance ( $dQ/dV$ ) plots of first cycle (inset) of  $B_{50}$  thin film/ $LiPF_6$ / $Li$  cell at (a) RT, (b)  $60^\circ\text{C}$ , (c)  $85^\circ\text{C}$ , (d) returned to RT between 0.01 V and 3.0 V vs.  $Li/Li^+$ .



**Fig. 3.** (a) B 1s and Li 1s XPS spectra of B thin film electrodes of the as-deposited and after discharging to 0.01 V. The *ex situ* HRTEM and SAED images of  $B_{50}$  thin film: (b) and (c) discharged to 0.01 V; (d) and (e) recharged to 3.0 V.

**Table 2**  
 $d$ -spacings ( $\text{\AA}$ ) derived from SAED analysis of as-deposited, first discharge to 0.01 V and recharge to 3.0 V of B thin film electrode with the referenced  $B_{50}$   $d$ -spacings are shown [17].

| As-deposited        | First discharge to 0.01 V | First charge to 3.0 V | $B_{50}$ ( $P4_2/nmm$ ) referenced [17] |
|---------------------|---------------------------|-----------------------|-----------------------------------------|
| 4.375               |                           | 4.348                 | 4.375 (1 0 1)                           |
| 3.949               |                           |                       | 3.917 (1 1 1)                           |
| 3.123               |                           |                       | 3.095 (2 1 1)                           |
| 2.763               |                           |                       | 2.767 (3 1 0)                           |
| 2.414               | 2.510                     | 2.414                 | 2.428 (3 1 1)                           |
| 1.987               |                           | 1.974                 | 1.957 (4 1 1)                           |
| 1.402               | 1.426                     | 1.406                 | 1.440 (3 1 3)                           |
| 1.356               |                           | 1.352                 | 1.367 (5 2 2)                           |
| $a = 8.75 \pm 0.05$ | $a = 9.06$                | $a = 8.72 \pm 0.05$   | $a = 8.75$                              |
| $c = 5.06 \pm 0.05$ | $c = 5.31$                | $c = 5.02 \pm 0.04$   | $c = 5.06$                              |



**Fig. 4.** (a) Total and local density of states (DOS) for tetragonal  $B_{50}$  and (b) energy dispersion relationship for tetragonal  $B_{50}$ . The Fermi level is set to zero.

space (appr.  $1.90 \text{ \AA}$  in radius) trapped by the four cages. Fig. 1(f) demonstrates that (i) the total energy of  $B_{50}$  lattice (black points) varies with its volume changes in a limited range (not larger than  $\pm 5.0\%$ ), and (ii) the red line corresponds to the fitting of the above relationship to the Birch–Murnaghan equation of state, where the parameters  $B_0$  and  $B'$  correspond to the bulk modulus and the differential of the  $B_0$  to the volume in Table 1, respectively [22]. The calculated lattice parameters,  $a = 8.851 \text{ \AA}$ ,  $c = 4.945 \text{ \AA}$  and thermodynamically equilibrated volume (only 0.002% expansion), are very close to the experimental values. A few reports on the dynamic properties of tetragonal B are found to evaluate the bulk modulus  $B_0$  (195.76 GPa),  $B_0$  is at the same level as 224 GPa for  $\alpha$  phase, 184 GPa for  $\beta$  phase and 22 GPa for  $\gamma$  phase of boron allotropes [1,2,23].

### 3.2. Electrochemical properties of tetragonal boron $B_{50}$

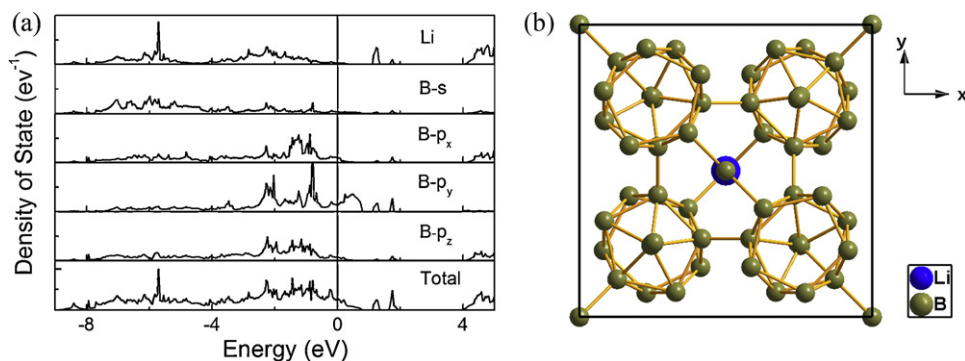
Fig. 2 shows the galvanostatic cycling curves and differential capacitance ( $dQ/dV$ ) plots at the first cycle (inset) of the tetragonal  $B_{50}$  thin film/Li cell at different conditions. The electrochemical performance of the swagelok-cell at RT is similar to pseudo-capacitor behavior with no obvious redox peaks in Fig. 2(a). Only  $44 \text{ mAh g}^{-1}$  is obtained, close to  $\text{LiB}_{50}$  ( $49 \text{ mAh g}^{-1}$ ).

Nevertheless, the electrochemical reactivity of the  $B_{50}$  thin film changes at elevated temperature. When the temperature rises to  $60^\circ\text{C}$  and  $85^\circ\text{C}$ , a slope-plateau around 1.0 V in the discharge/charge curves (a remarkable redox peak in  $dQ/dV$  plot) can be seen in Fig. 2b and c. The capacity increases rapidly, about  $192 \text{ mAh g}^{-1}$  at  $60^\circ\text{C}$ , close to  $\text{LiB}_{12}$  ( $196 \text{ mAh g}^{-1}$ ) and  $268 \text{ mAh g}^{-1}$  ( $\sim \text{LiB}_9$ ) at  $85^\circ\text{C}$ . The appearance of the plateau means that lithium could be stored in the  $B_{50}$  structure and may break down the boron icosahedra of the tetragonal boron lattice. It is noticed that the

capacity is decreased to  $65 \text{ mAh g}^{-1}$  after returning to room temperature, slightly higher than  $44 \text{ mAh g}^{-1}$  in Fig. 2(a). This indicates that thermal activation process may change the microstructure irreversibly and increase the electrochemical activity. Due to the limited thermal stability of current nonaqueous electrolyte, it is not possible to test thermal activation at higher temperature, such as  $400\text{--}500^\circ\text{C}$ . This will be checked in future using proper solid state electrolyte.

Fig. 3(a) shows B 1s and Li 1s XPS spectra of thin film electrodes of the as-deposited and of the discharging to 0.01 V. The B 1s peak shifts slightly from 191.2 to 190.1 eV. The Li 1s peak at 55.0 eV appears after discharging to 0.01 V. This indicates the formation of Li–B compound since the binding energy does not belong to any Li–O or Li–F species of SEI layer and the electrolyte [24]. When the film discharged to 0.01 V at RT, the SAED pattern in this phase shown in Fig. 3(c) consists of two obvious rings. The  $d$ -spacings derived from these two rings have slightly increased from  $2.414 \text{ \AA}$  to  $2.510 \text{ \AA}$  (plane (3 1 1)) and from  $1.356 \text{ \AA}$  to  $1.426 \text{ \AA}$  (plane (3 1 3)) respectively, which could be ascribed to the Li insertion into the boron lattice. After recharging to 3.0 V, some discrete spots appear in the SAED pattern as shown in Fig. 3(e), which can be indexed to the tetragonal  $B_{50}$ , indicating the reversible Li insertion/extraction into/from the boron lattice.

Table 2 summarizes the  $d$ -spacings derived from SAED patterns of as-deposited, discharged to 0.01 V, and recharged to 3.0 V of the  $B_{50}$  thin-film electrode, which clearly show the variations of the planar distances. Since only the pattern of the  $B_{50}$  ( $d$ -spacings consistent well with  $B_{50}$ ) is found in the SEAD patterns, it indicates the solid-solution reaction upon Li insertion/extraction (Li–B alloying reaction). Based on these results, the electrochemical reaction mechanism of tetragonal boron anode



**Fig. 5.** (a) Shows the total and local DOS for tetragonal  $\text{LiB}_{50}$ . The Fermi level is set to zero. (b) Demonstrates the  $\text{LiB}_{50}$  lattice, where the Li ion locates at Li1 (0.50, 0.50, 0.50).



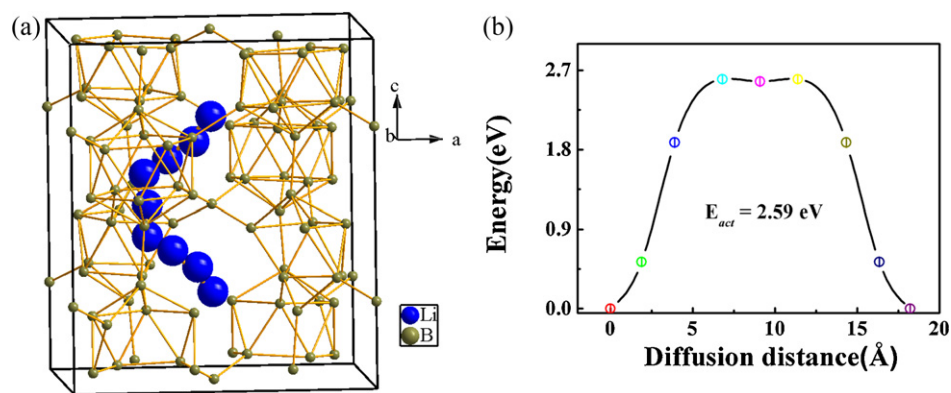
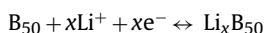


Fig. 6. (a) Spatial Li ion diffusion path in a  $1 \times 1 \times 2$   $B_{50}$  superlattice and (b) the corresponding activation energy for Li ion diffusion (eV).

at shallow lithium insertion level could be initially proposed as follows:



### 3.3. Electronic structures of tetragonal $B_{50}$

Fig. 4(a) and (b) display the density of states (DOS) and band structure of tetragonal boron ( $B_{50}$ ), respectively. From Fig. 4(a), the fully symmetric spin up and spin down DOS indicates the non-spin-polarization in the tetragonal  $B_{50}$  system and further implies its metallic characteristic which is much different from the other B phases [1–2], because the Fermi level goes across the DOS at 0 eV. Moreover, it can be deduced that the  $B p_x$  orbital contributes most to the valence electrons and the B s orbitals hybrid with the p orbitals obviously. Besides its metallic property in Fig. 4(b), the energy dispersions of the electrons near Fermi level are ca. 2 eV, indicating its de-localization trend, plausibly implying the good electronic conductivity of tetragonal  $B_{50}$ . This will be checked in future.

### 3.4. Calculation on lithium storage in tetragonal $B_{50}$

After insertion of one Li into the tetragonal  $B_{50}$  lattice to form the  $LiB_{50}$  phase, a 0.7% volume expansion with  $a = 8.879$  Å and  $c = 4.951$  Å is found based on the calculation. The Li ion stays at Li1 (0.50, 0.50, 0.50) site after fully relaxation as shown in Fig. 5(b). Fig. 5(a) reveals the total and local DOS of the tetragonal  $LiB_{50}$ . Although the B p orbitals still dominate the valence electrons, the main contributions come from the  $p_y$  orbital of  $LiB_{50}$ , not the same as  $p_x$  in the  $B_{50}$  lattice. From the LDOS of Li in Fig. 5a, it reveals that little contribution from the Li to the valence electrons is devoted. Moreover, the residue charge near the inserted Li can be treated as zero from the spatial charge integration at a radius equal to 0.76 Å, indicating the Li is ionized. Further Bader atomic charge [25,26] analyses for B ions show that the charge distribution of B ion ranges from 2.875( $e^-$ ) to 3.161( $e^-$ ), this implying that firstly B ion obtains the electron from Li and then the electron might tend to behave as itinerant role to enhance the electronic conductivity.

The nudged elastic band (NEB) method [27] is employed to calculate the activation energy and simulate migration paths for Li ion migration in the  $B_{50}$  supercell. Activation energy roughly refers to the energy barriers that the Li ion has to overcome on jumping from one site to an adjacent vacant site. Fig. 6(a) and (b) demonstrates the Li ion diffusion from Li1 (0.50, 0.50, 0.25) to Li2 (0.50, 0.50, 0.75) in a  $1 \times 1 \times 2$   $B_{50}$  supercell. A high 2.59 eV energy barrier indicates the poor Li ion diffusion kinetics at absolute zero. It is found that the nearest B ion shifts far away, meaning the break of the  $B_{12}$  icosahedra when the Li ion goes across the

narrowest parts (1.65 Å in Fig. 1(a)) of two adjacent cages. The higher energy barrier also explains the poor capacity obtained at RT in Fig. 2(a), while the poor capacity should be improved at elevated temperature.

## 4. Conclusion

A polycrystallized tetragonal  $B_{50}$  thin film has been fabricated by reactive PLD method in vacuum chamber and tested as Li storage electrode for the first time. The obtained  $B_{50}$  thin film shows a lithium storage capacity of only  $44 \text{ mAh g}^{-1}$  between 0.01 V and 3.0 V in a nonaqueous electrolyte at room temperature. First-principles calculation indicates that the  $B_{50}$  is a metallic conductor, but a poor lithium ionic conductor,  $\sim 2.59$  eV energy barrier for Li ion hopping at absolute zero, which explains its low RT reactivity toward lithium. The lithium storage capacity increases to  $192 \text{ mAh g}^{-1}$  at  $60^\circ\text{C}$ , and  $268 \text{ mAh g}^{-1}$  at  $85^\circ\text{C}$ . This could be caused by enhanced ionic conductivity at elevated temperature, needing further clarification.

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